SPH6004 Individual Assignment Report

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1. Introduction

The Medical Information in Critical Care Market IV (MIMIC-IV, Version 3.1) is a large-scale, publicly accessible electronic health record (EHR) database [1]. This project utilizes structured clinical features extracted from MIMIC-IV records of critically ill patients to construct predictive models assessing the likelihood of patients leaving the ICU.

This study aims to: (1) develop predictive models based on structured clinical data; (2) employ feature selection methods to reduce dimensionality and enhance model robustness; (3) compare and analyze performance differences among various predictive models. Through systematic experimentation, we evaluate how feature selection strategies and model selection influence the discriminatory power and generalization performance of ICU discharge prediction.

2. Methodology

2.1 Exploratory Data Analysis (EDA)

An exploratory data analysis (EDA) was conducted to understand the structure and quality of the dataset before preprocessing. The analysis examined the dataset size, feature types (numerical, categorical, and binary), and the proportion of missing values across all variables. Columns with substantial missingness were identified to guide subsequent data cleaning decisions. In addition, the distribution of the target variable 'icu_death_flag' was analyzed to assess potential class imbalance in ICU mortality outcomes.

2.2 Data Preprocessing

2.2.1 Data Cleaning and Missing Value Filtering

To improve data quality and reduce noise caused by excessive missing values, the proportion of missing values was calculated for each feature. Features with a missing rate greater than 50% were removed from the dataset.

2.2.2 Dataset Split

The dataset was divided into training and testing sets using a 7:3 ratio. To maintain the

original distribution of the target variable icu_death_flag, stratified sampling was applied. After splitting, the indices of both datasets were reset to ensure correct alignment between features and labels during subsequent preprocessing steps.

2.2.3 Missing Value Imputation

Patient demographic information included gender, race, marital status, first care unit, insurance type, and language. Missing values in these categorical variables were replaced with the label "Unknown" to preserve record completeness without removing samples. Missing values in numerical variables were imputed using the median of the training data to reduce the influence of outliers. 'gcs_unable' was a special binary indicator variable, which was imputed using the mode. The imputation models were fitted only on the training set and subsequently applied to the test set to prevent data leakage.

2.2.4 Standardization

To ensure that numerical features are on a comparable scale during model training, Z-score standardization was applied to all numerical variables. The scaler was fitted only on the training data and subsequently applied to both the training and testing sets. This approach prevents information from the test set from influencing the training process and helps maintain proper model evaluation.

2.2.5 Categorical Encoding

Categorical variables were converted into a numerical format using One-Hot Encoding [2]. This method creates binary indicator variables for each category level. To ensure that the training and testing datasets have consistent feature dimensions, the encoded test set was aligned with the training set by adding missing columns and filling them with zeros.

2.3 Feature Selection

To reduce model complexity and enhance interpretability, we employ L1 regularized logistic regression for feature selection. L1 regularization promotes model sparsity by introducing a penalty term on the absolute values of model coefficients [3]. Consequently, feature coefficients with low information content are precisely reduced to zero, effectively eliminating these features. This property makes L1 logistic regression particularly suitable for high-dimensional clinical datasets, where numerous variables may be redundant or possess weak predictive power. Additionally, the prediction task in this study is binary classification. Logistic regression provides probabilistic interpretations and allows direct examination of feature coefficients, thereby enhancing the interpretability of selected predictor variables [3].

To determine the optimal regularization strength, a regularization path experiment was conducted by evaluating several candidate values for the hyperparameter C, where $C = 1/\lambda$ and λ represents the regularization strength. Smaller C values correspond to stronger regularization effects, resulting in fewer selected features [3].

C = [0.005, 0.01, 0.02, 0.03, 0.04, 0.05]. For each C value, an L1-regularized logistic regression model was trained using the liblinear solver. Due to severe dataset imbalance, the regularization weight for each class was adjusted during training via the parameter class_weight = 'balanced'.

Model performance was evaluated using 5-fold stratified cross-validation, with the AUC as the metric. Stratified cross-validation ensured each fold retained the original class distribution of the target variable.

2.4 Hyperparameter Tuning and Modeling

To optimize model performance, a grid search approach with cross-validation was employed to tune hyperparameters across all machine learning models, including logistic regression [3], decision trees [4], random forests [5], AdaBoost [6], XGBoost [7], and SVM [8]. Predefined parameter grids were constructed for each model, with optimal parameter combinations selected based on the highest AUC scores achieved during cross-validation. A 5-fold stratified cross-validation strategy was adopted to ensure balanced class distribution throughout the tuning process. Given the severe class imbalance in the dataset, the regularization weight for each class was adjusted using the parameter `class_weight = 'balanced'`.

After hyperparameter tuning, each model was retrained on the full training set using its optimal configuration. For XGBoost, an early stopping mechanism was additionally applied: training was terminated if the validation set AUC failed to improve for 30 consecutive iterations, and the model was fitted using the final number of estimators determined as optimal. All models were ultimately evaluated on an independent test set. Performance assessment utilized six metrics: accuracy, ROC-AUC, precision, recall, F1 score, and macro F1 score.

The code for this project has been uploaded to GitHub (https://github.com/Wu-Bohan/SPH6004-Individual-Assignment).

3. Result

3.1 Dataset Description

This raw dataset comprises 65,366 ICU admission cases, featuring a total of 140 characteristics. These include 127 numerical variables, 9 categorical variables, and 4 binary variables. The dataset exhibits substantial missing values. A total of 129 features contain missing data, with 88 features having a missing rate exceeding 50%. The target variable 'icu_death_flag' indicates whether a patient died during their ICU stay. Its category distribution is severely imbalanced: 91.32% of cases survived, while only 8.68% died. This imbalance necessitates particular attention during model training to avoid bias toward the majority class.

After applying the missing value filtering procedure, 88 features with high missing rates were removed from the dataset, ultimately retaining 43 features for subsequent analysis. The dataset was split into training and testing sets using a 7:3 stratified split. Missing values in categorical variables were replaced with the label "unknown," while numerical variables were imputed using the median of the training data. Subsequently, numerical features were standardized using Z-score normalization, while categorical variables were converted via one-hot encoding. The final feature matrix comprised standardized numerical features and encoded categorical variables, totaling 123 features.

The training set comprises 45,755 samples, including 3,970 fatal cases (8.68%). The test set contains 19,610 samples, including 1,701 fatal cases (8.67%).

3.2 Feature Selection

L1 logistic regression was used for feature selection. Based on cross-validation results, C = 0.01 was selected as the optimal regularization parameter (Table 1). This value achieves an effective balance between model performance and feature sparsity, maintaining competitive predictive capability while reducing the number of features utilized. Using this optimal parameter, the final L1 logistic regression model was trained on the complete training dataset.

Table 1. Result of L1 Regularization Path

C	Number of Selected Features	Mean CV AUC
0.005	31	0.9001
0.01	37	0.9030
0.02	46	0.9041
0.03	51	0.9044
0.04	55	0.9047
0.05	58	0.9048

After model training completion, features with non-zero coefficients were retained as predictive variables, while those with coefficients shrunk to zero were removed from the model. The magnitude of the coefficient absolute value reflects the strength of the feature's

correlation with the risk of ICU mortality.

Ultimately, 37 features were selected from the original 123, achieving significant dimensionality reduction while preserving predictive information. The final selected feature set was used to construct a streamlined training and testing dataset to support subsequent model development.

3.3 Model Performance

3.3.1 Hyperparameter Tuning

Before model training, grid search was performed to optimize the hyperparameters of each model. The best hyperparameter settings are summarized in Table 2.

Model	Best ROC-AUC (CV)	Optimal Hyperparameters
Logistic Regression	0.9046	C = 0.1, penalty = 'l2', solver = 'lbfgs'
Decision Tree	0.8679	criterion = 'entropy', max_depth = 7, min_samples_split = 2, min_samples_leaf = 20, max_features = None, ccp_alpha = 0.001
Random Forest	0.9096	n_estimators = 300, criterion = 'entropy', max_depth = 10, min_samples_split = 2, min_samples_leaf = 20, max_features = sqrt, ccp_alpha = 0.0
AdaBoost	0.9162	base_max_depth = 2, n_estimators = 200, learning_rate = 0.1, algorithm = 'SAMME.R'
XGBoost	0.9197	colsample_bytree = 0.8, gamma = 0, learning_rate = 0.05, max_depth = 5, min_child_weight = 10, n_estimators = 200, reg_alpha = 0, reg_lambda = 1.0, subsample = 0.8
SVM	0.9130	C = 0.1, Gamma = 'auto', Kernel = 'rbf'

3.3.2 Model Performance

Among all six models, XGBoost demonstrated the strongest overall performance, ranking first in ROC-AUC (0.9235), F1 score (0.5034), and accuracy (0.8603), establishing itself as the optimal model for predicting ICU mortality in this study (Figure 1).

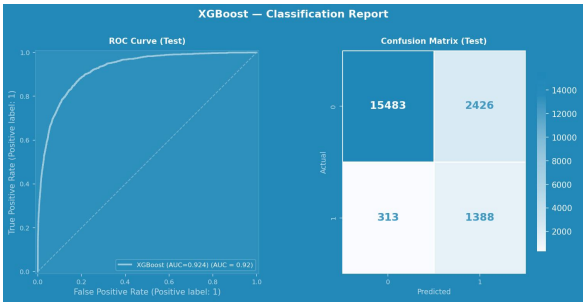

Figure 1. ROC Curve and Confusion Matrix of the XGBoost Model on the Test Set

The ROC-AUC value of 0.9235 indicates that XGBoost reliably distinguishes between surviving and deceased patients across all classification thresholds. Regarding sensitivity, XGBoost achieved a recall rate of 0.816, correctly identifying 81.6% of actual ICU deaths. XGBoost achieved the optimal balance between precision and recall, with its F1 score of 0.503 ranking first. This result primarily stems from XGBoost's gradient boosting framework, which iteratively corrects prediction errors and utilizes regularization to control overfitting [7]. Concurrently, the scale_pos_weight parameter enhances recognition capabilities for the minority class.

Table 3. Performance Comparison of Machine Learning Models on the Test Set

Model	**Accuracy**	**ROC-AUC**	**Precision**	**Recall**	**F1-Score**	**Macro F1**
Logistic Regression	0.838	0.9058	0.3264	0.816	0.4663	0.6854
Decision Tree	0.7482	0.8669	0.2335	0.8336	0.3649	0.6039
Random Forest	0.868	0.9119	0.3723	0.7613	0.5001	0.712
AdaBoost	0.8569	0.9206	0.3573	0.8136	0.4965	0.7065
XGBoost	0.8603	0.9235	0.3639	0.816	0.5034	0.711
SVM	0.8348	0.9157	0.3257	0.8454	0.4702	0.6862

Table 3 summarizes the predictive performance of six machine learning models on the test dataset. Model performance was evaluated using accuracy, ROC-AUC, precision, recall, F1 score, and macro F1 score.

Among all models, XGBoost achieved the highest AUC (0.9235), indicating its strongest overall capability in distinguishing ICU survivors from non-survivors. AdaBoost (AUC=0.9206) and SVM (AUC=0.9157) also demonstrated robust discrimination capabilities, with AUC values surpassing both logistic regression and tree-based models.

Regarding accuracy, Random Forest ranked first at 0.868, followed by XGBoost (0.8603) and AdaBoost (0.8569). However, due to the extremely high proportion of survivors in the dataset causing severe class imbalance, accuracy alone cannot fully reflect model performance.

There are significant differences in recall across models. SVM achieved the highest recall (0.8454), followed by Decision Tree (0.8336) and XGBoost (0.8160), demonstrating superior performance in identifying patients at high risk of death. In contrast, Random Forest exhibited a lower recall rate (0.7613), indicating this model was more prone to missing death cases compared to the other approaches.

Regarding precision, all six models recorded notably low values, with scores ranging from 0.2335 to 0.3723. Decision Tree has the lowest precision (0.2335), indicating that the majority of its positive predictions were false alarms. This pattern is consistent across all models and reflects the difficulty of achieving high precision under severe class imbalance, where the class-balancing strategy prioritises recall at the expense of precision.

Overall, XGBoost achieves the optimal balance between discriminative power and minority class detection, attaining the highest ROC-AUC value while maintaining high recall performance. Boosting-based models generally outperform traditional models in identifying ICU mortality cases, likely due to their ability to capture complex nonlinear relationships within clinical data.

4. Discussion

L1-regularized logistic regression demonstrated exceptional performance in dimensionality reduction, streamlining the initial 123 features into 37 key predictors. By selecting C = 0.01 as the optimal regularization parameter, the model significantly enhanced sparsity while maintaining a high cross-validation AUC of 0.9030.

Comparative analysis of six machine learning architectures revealed that boosting-based ensemble models exhibited strong predictive capabilities. XGBoost emerged as the optimal model, achieving the highest ROC-AUC (0.9235) and F1 score (0.5034) on the test set. Its success may be attributed to the gradient boosting framework and the application of the `scale_pos_weight` parameter, which effectively addressed the issue of low mortality cases by increasing the cost of misclassifying minority errors [7].

All six models achieved recall rates exceeding 0.7. In clinical settings, high recall is typically prioritized, as the cost of false negatives far outweighs that of false positives.

The primary challenge faced in this study was severe class imbalance, with fatal cases

constituting only 8.68% of the dataset. Under these conditions, models often generate more false positives while attempting to maximize recall and identify fatal cases, leading to decreased precision. This outcome highlights the common "precision-recall tradeoff" phenomenon in imbalanced datasets [9].

Nearly all models achieved high ROC-AUC values, indicating that despite lower precision, the models possess strong inherent capabilities for correctly identifying high-risk patients.

5. Conclusion

This study developed multiple machine learning models based on clinical data to predict mortality in intensive care units. After data preprocessing and feature selection using L1-regularized logistic regression, 37 key features were retained for model training. Hyperparameters for each model were optimized through grid search with five-fold cross-validation.

Comparing the performance of logistic regression, decision trees, random forests, AdaBoost, XGBoost, and support vector machines (SVM), XGBoost demonstrated optimal overall performance with the highest ROC-AUC value (0.9235), indicating strong discriminative capability. Boosting-based models also exhibited high recall in identifying fatal cases, which is particularly crucial for handling imbalanced clinical datasets.

Future research may explore other techniques for handling class imbalance, such as resampling methods or cost-sensitive learning, to mitigate class imbalance issues and enhance model performance.

References:

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Appendix

GitHub: https://github.com/Wu-Bohan/SPH6004-Individual-Assignment

Large Language Models (LLMs) were used to assist with parts of this project. Claude was used to assist with code development, and ChatGPT-5.0 was used to improve the clarity and language of the report.